# A1.1 Water (HL)

NAME

DATE

**How to use this sheet:** go through the slides and videos with it, then close them and write from memory. Use your own words and add page numbers from the book (pp. 11–13) where you want more detail.

**Guiding questions.** Which features of water make life possible? What does living in water make easier and harder for organisms?

## A1.1.7 Where Earth's water came from

Why could the young Earth not condense water? Where could water ice form instead?

Evidence for the asteroid hypothesis (include deuterium):

Two reasons Earth kept its water. Why did Mars lose most of its surface water?

## A1.1.8 Water and the search for life

Sketch the Sun's habitable zone, marking Venus, Earth and Mars.

How do scientists detect water on distant planets? Why is a planet in the habitable zone not certain to have life?

## Before you finish

Tick the ones you can explain without notes. Circle anything to ask about.

- [ ] Extraplanetary water
- [ ] Asteroids
- [ ] Carbonaceous chondrites
- [ ] Deuterium
- [ ] Gravity
- [ ] Condensation
- [ ] Goldilocks zone
- [ ] Transit spectroscopy